

Power Consumption Analysis of Terrestrial Systems with the Same Fiber Capacity

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Abstract We analyze the relative power consumption of terrestrial long-haul systems constructed with different fiber types and optimized span lengths producing the same cable capacity. Advanced fibers with C-band amplifiers have the lowest power consumption and carbon footprint, saving up to 40% over conventional fiber. ©2026 The Author(s)

Introduction

Standard single-mode fiber compatible with the G.652.D ITU standard has long been the most widely deployed optical fiber in terrestrial long-haul systems. In recent years, new fibers with improved transmission properties such as ultra-low attenuation and/or larger fiber effective area have been introduced that comply with other standards such as G.654.C and G.654.E [1-3]. These advanced fiber types enable longer transmission distances, longer span length, or greater fiber capacity depending on the transmission system design. In [4], G.654.E fiber with ultra-low attenuation and large effective area enabled real-time 64 x 1 Tb/s transmission over 1260 km with variable ultra-long spans up to 160 km. Lately, power consumption in amplifier huts has become a critical system issue as cable fiber counts and number of parallel ducts increase. With this in mind, it is important to evaluate different fiber types and amplifier types for power consumption in greenfield systems during system design.

In this paper, we perform a power consumption analysis for an example long-haul terrestrial system. We assess both C-band only (denoted C+C) and C+L systems, each for four fiber types. We also consider systems using only erbium-doped fiber amplifiers (EDFAs) and systems employing hybrid Raman/EDFA amplifiers. Our approach is to find the maximum span lengths for each fiber and system type that produce the same cable capacity. This is done by determining the span lengths that deliver the same

generalized signal-to-noise ratio (GSNR) and received SNR over the link length and requiring cables with twice the fiber count for C+C systems as C+L systems. The span lengths for each fiber and system type are generally different leading to variations in numbers of amplifiers in the link. With representative power consumption values for C- and L-band EDFAs, as well as hybrid Raman/EDFAs of both types, we can then compare the relative power efficiency of each fiber and system type. We find that advanced fibers with lower attenuation and larger effective area that enable longer span lengths also yield the lowest power consumption by up to 40% compared to standard single-mode fiber (G.652.D). Moreover, hybrid Raman/EDFA amplified systems require the most power overall even though they allow the longest span lengths. Overall, our results suggest that G.654.E fiber with C-band only amplifiers is the most power efficient terrestrial long-haul system.

Transmission System Modeling

The system studied for this analysis was a long-haul terrestrial system with nominal length of 2000 km. We considered four optical fibers with characteristics and corresponding ITU standard listed in Table 1.

We considered all-EDFA and hybrid Raman/EDFA amplification for both C+C and C+L systems. We modeled GSNR as a function of span length using an inter-channel stimulated Raman scattering- Gaussian Noise (ISRS-GN)

Table 1: Terrestrial optical fibers considered

Parameter	Fiber 1 (G.652.D)	Fiber 2 (G.654.C)	Fiber 3 (G.654.E)	Fiber 4 (G.654.E)
Attenuation (dB/km)	0.183	0.158	0.166	0.150
Effective area (μm^2)	80	83	125	125
Dispersion (ps/nm/km)	18	18	21	21
n_2 ($\times 10^{-20} \text{ m}^2/\text{W}$)	2.3	2.1	2.2	2.1

[5,6] model of nonlinear fiber propagation, including channel dependent attenuation and Raman gain coefficients of the fibers to account for span-length increasing tilt due to ISRS. The launch power was optimized for each fiber type and span length to maximize the total fiber capacity with individual channel powers being optimization variables. Four Raman pumps were employed for C-band systems (maximum total pump power limit of 1 W, maximum individual pump limit of 0.5 W) and 5 Raman pumps were employed for C+L systems (total pump limit of 1.2 W). The EDFA total output power limit was 23 dBm.

The results for average GSNR as a function of span length for the 2000 km link are shown in Fig. 1 for (a) C-band and (b) C+L EDFA-only systems. The span length for the baseline G.652.D Fiber 1 was set to 80 km for EDFA-only C+C and C+L, and the span lengths were determined for the other fibers and all amplifier configurations to match the GSNR of Fiber 1 at 80 km. In general, the curves for Fibers 2 and 3 intersect in the range of 120-150 km. This is due to the EDFA output power limit causing Fiber 3 to operate in the linear regime starting at span lengths about 20 km shorter than other fibers. Another effect is the smaller Raman gains of Fibers 3 and 4 by about 5 dB than those of Fibers 1 and 2 because of the pump power limits.

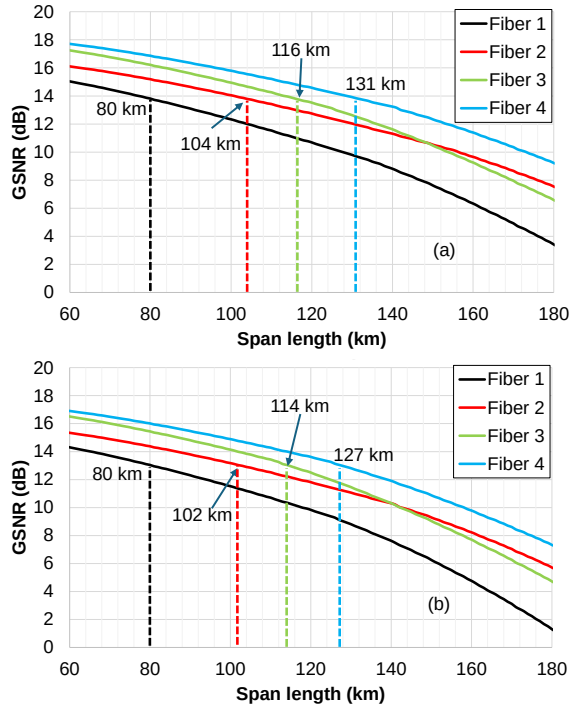


Fig. 1: Average GSNR vs. span length for EDFA-only systems: (a) C-band, and (b) C+L.

The general system parameters are shown in Table 2 and a summary of the span lengths for all fibers in all four system types is given in Fig. 2. The average GSNR value for all C+C systems

was 13.8 dB while it was 13.0 dB for all C+L systems, leading to average fiber capacities of 30 and 56 Gb/s, respectively, according to Eq. 1. GSNR was calculated including 2 dB loss margin as well as nominal losses for splices, connectors, C/L bandsplitters, and Raman pump combiners. The C+C and C+L fiber counts per cable were 288 and 144, respectively, to produce essentially equivalent cable capacities within a few percent. The amplifier power consumption values represent budgeted power including commons.

$$C_{\text{fiber}} = 2BN_{\text{ch}} \log_2 \left[1 + \frac{(10^{-(\text{GSNR})/10} + 10^{-(\text{SNR}_m)/10})^{-1}}{10^{(\Gamma/10)}} \right] \quad (1)$$

Table 2: General System Parameters

Parameter	Value
Nominal Link length (km)	2000
C-band NF (dB)	5
L-band NF (dB)	6
C/L bandsplitter loss (dB)	0.5
Baud rate (Gbaud)	135
Channel spacing (GHz)	150
Number of channels per band	30
Transceiver SNR _m (dB)	19
Gap-to-Shannon Γ (dB)	2
C-band EDFA power (W)	67
C+L-band EDFA power (W)	161
Hybrid Raman/EDFA C+C power (W)	156
Hybrid Raman/EDFA C+L power (W)	300
C+C cable fiber count	288
C+L cable fiber count	144

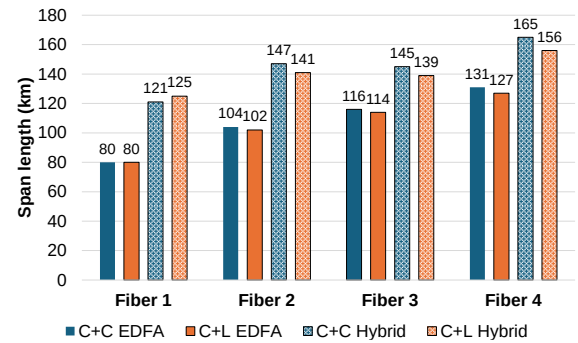


Fig. 2: Span lengths for all fibers and system configurations.

Power Consumption Results

Using the span lengths illustrated in Fig. 2, we counted the number of amplifiers for each fiber and system type to assess the total link power consumption of each system. Recall that for C+C systems, the cable fiber count was assumed to be 288, while the fiber count was 144 for C+L systems to ensure comparable cable capacities for all systems. A summary of the total link power consumption for all systems is shown in Fig. 3.

The results are normalized to the Fiber 1 C+C EDFA-only system for easy overall comparison.

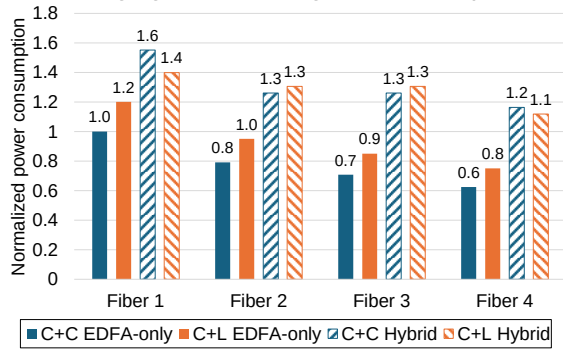


Fig. 3: Summary of normalized power consumption of all system types and fibers.

Several points can be made by observation of the results in Fig. 3. The first is that for EDFA-only systems, C+L systems consume more power than C+C systems for all fibers because of the higher power draw of L-band EDFAs compared to C-band EDFAs. A second point to note is that Fibers 2, 3, and 4 all enable lower power consumption with EDFA-only amplifiers than the reference system of Fiber 1 (C+C, EDFA-only). Even EDFA-only C+L systems with Fibers 2, 3, and 4 require lower power than Fiber 1 C+C. This is because of the longer span lengths of Fibers 2, 3, and 4 compared to Fiber 1. Another observation from Fig. 3 is that all fibers using hybrid Raman/EDFA amplifiers in either C+C or C+L configurations consume more power than the reference system by about 12-55%, even though they enable the longest span lengths. However, the relative power consumption of hybrid C+C and hybrid C+L systems depended somewhat on the fiber type, unlike the EDFA-only systems.

While Fig. 3 conveys a summary of the power consumption results, it can be interesting to put them in the context of cost savings and reduced carbon dioxide (CO₂) emissions to the atmosphere over a given time period. However, since the Hybrid Raman/EDFA systems (C+C and C+L) required more power for all fiber types than the EDFA-only systems, we confine the analysis to the EDFA-only systems. For these comparisons within that group, we use a baseline system of Fiber 1 C+L as the reference because it consumes the most power. Results for savings in the cost of power over a 10-year period for all fibers and C+C and C+L systems relative to the baseline case are shown in Fig. 4. The cost of electricity was assumed to be \$0.30/kW-hr for these results. Expressed in this manner, Fiber 1 C+C and all other fibers with either C+C or C+L systems enable cost savings from power consumption relative to the baseline. Fiber 4 C+C with the longest span length would save more than \$7M over 10 years relative to the baseline Fiber 1 C+L.

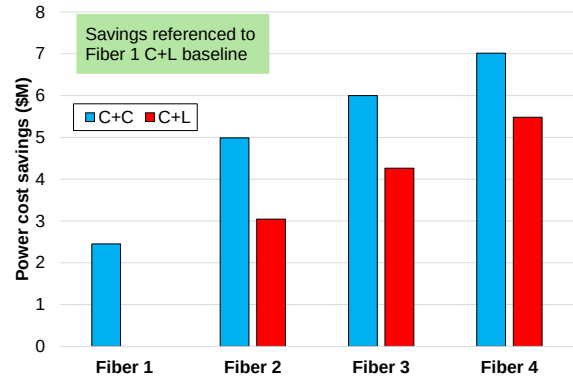


Fig. 4: Power cost savings of all fibers over a 10-year period relative to the baseline Fiber 1 C+L system.

A similar analysis calculates the reduction in CO₂ emissions over the same 10-year period relative to the baseline case. These results are illustrated in Fig. 5. An average value of 0.37 kg of CO₂ emitted per kW-hr was used to calculate these results. Fig. 5 shows that all fibers and systems enable CO₂ emission reductions relative to the baseline, up to more than 8600 metric tons for Fiber 4 C+C.

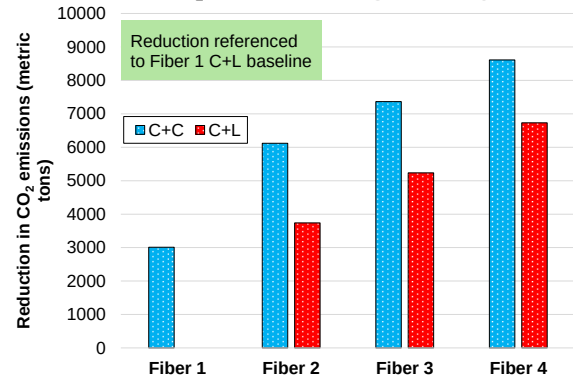


Fig. 5: Reduction of CO₂ emissions of all fibers over a 10-year period relative to the baseline Fiber 1 C+L system.

Conclusions

We have analysed green-field terrestrial system construction for different fiber types and amplification configurations in terms of relative total link power consumption. The total cable capacity for each was kept essentially the same and maximum span lengths for each fiber and system type were determined that produced the same received GSNR. A G.654.E fiber (Fiber 4) produced the greatest power consumption savings of close to 40% relative to the baseline fiber because it enabled the longest span lengths due to ultra-low attenuation and large effective area. We also found that both C-band and C+L systems (EDFA-only) with all of Fibers 2, 3, and 4 produced lower power consumption than the G.652.D Fiber 1. Hybrid Raman/EDFA systems consumed the greatest power despite enabling longer span lengths than EDFA-only systems.

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